SOFTWARE ENGINEERING LABORATORY

LAB–4

ACTIVITY DIAGRAM

LIBRARY MANAGEMENT SYSTEM

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Course	B.Tech – Computer Science and Engineering
Subject	Software Engineering
Assignment	Lab–4: Draw the Activity Diagram
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UML Behavioural Modelling

1. Aim

To draw an activity diagram for the Library Management System showing the sequence of activities, decisions, and alternate flows involved when a student searches for and requests a book.

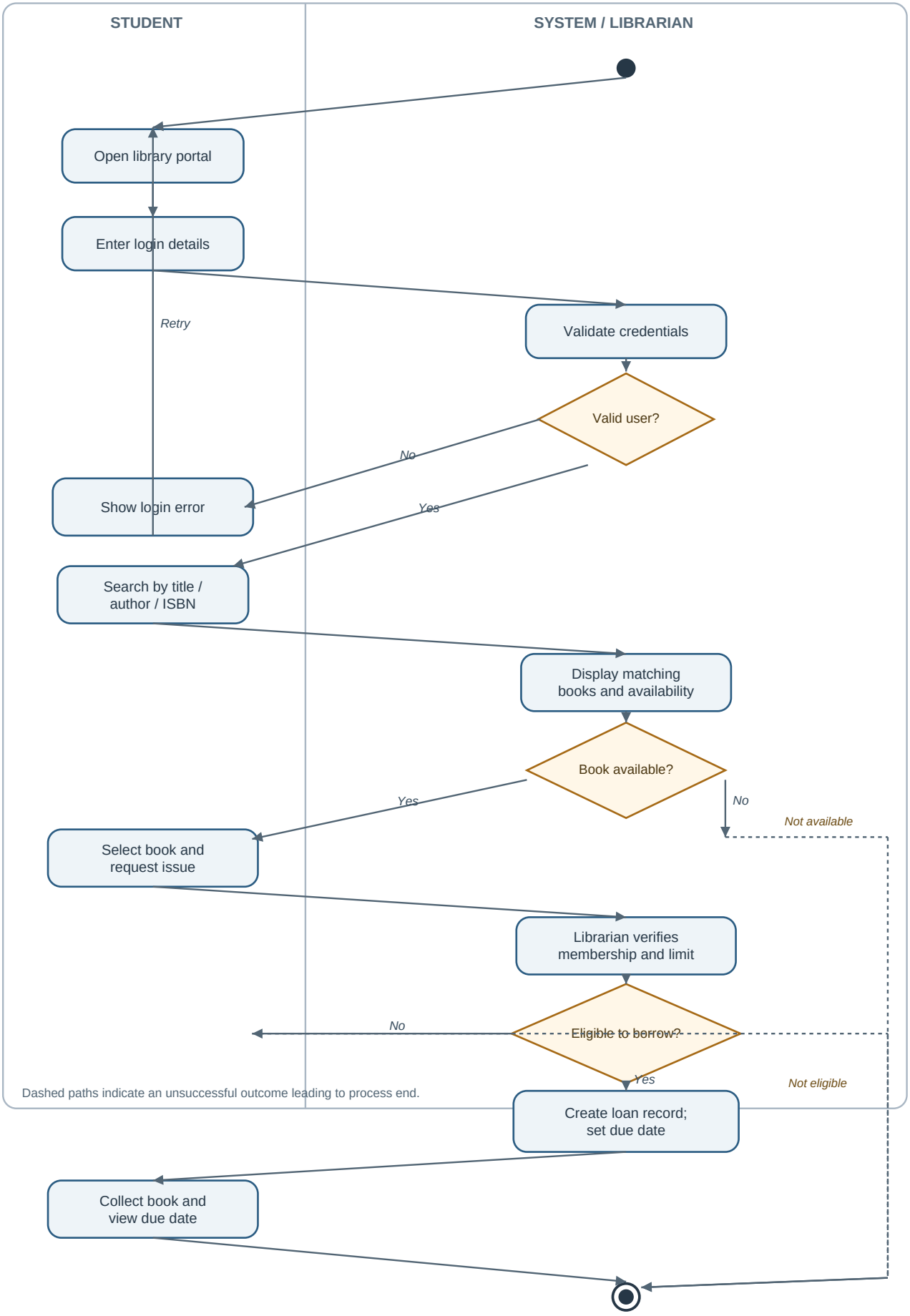
2. Introduction

An activity diagram is a UML behavioural diagram used to represent the workflow of a system. It shows the flow of control from one activity to another, including decision points and alternative paths. It helps in understanding how users and the system interact during a process.

3. Activity Diagram

The diagram below models the book-searching and book-issuing workflow. It includes login validation, book availability checking, librarian verification, loan creation, and unsuccessful paths when the user is invalid, the book is unavailable, or the student is not eligible to borrow.

ACTIVITY DIAGRAM – LIBRARY MANAGEMENT SYSTEM



4. Explanation of Activities

Step	Activity / Decision	Description
1	Open library portal	The student opens the Library Management System.
2	Enter login details	The student enters registered credentials.
3	Validate credentials	The system checks the entered credentials.
4	Valid user?	If no, an error is displayed and the student can retry. If yes, the process continues.
5	Search for a book	The student searches using title, author, or ISBN.
6	Display matching books	The system displays matching records and availability.
7	Book available?	If no, the request ends. If yes, the student selects the book.
8	Librarian verification	The librarian checks membership validity and borrowing limits.
9	Eligible to borrow?	If no, the request is rejected. If yes, the system creates the loan record.
10	Create loan record	The system stores the issue date and due date and updates available quantity.
11	Collect book	The student collects the book and can view the due date.

5. Conclusion

The activity diagram represents the complete workflow for searching and issuing a book in the Library Management System. It shows normal execution as well as alternate paths for invalid login, unavailable books, and ineligible borrowing requests. This diagram helps developers and testers understand the process flow and identify the system's decision points.

END OF LAB–4